

Date: / /

Pdf of book  
Tables of photocopy  
formula sheet  
calculator

Prob { Stats.

## STATISTICS

- Collecting the data
  - Organising the data
  - Summarizing the data
  - presenting the data
  - Analysing the data.
- } to get meaningful info.

Variable  $\rightarrow$  Temperature  $^{\circ}\text{C}$

Data 28, 29, 30, 27, 28, 32, 33, 34

Quantitative 25, 27, 26, 28, 29, 30, 31, 32

Qualitative 24, 26, 27, 30.

Sample size = 20

## Frequency Distribution.

(data should be according to your interest.)

- Minimum value of data = 24
- Max value of data. = 34
- Range = max - min =  $34 - 24 = 10$
- $k = \text{No. of intervals} = 1^k + 3.33 \log(n)$



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Date: / / Construct a frequency table!

$$= 1 + 3.33 \log(20)$$

$$= 5.5$$

$$= 6.0$$

4 Class Width =  $\frac{\text{Max} - \text{Min}}{K} = \frac{34 - 24}{6} = 1.66 \approx 2$

Class boundaries

Tally marks

24 and less than 26 11

26 and less than 28 1111

28 and less than 30 1111

30 and less than 32 1111

32 and less than 34 111

34 and less than 36 1

Frequency

Relative freq

% Cumulative

$$f_1 = 2$$

$$\frac{2}{20} = 0.1$$

$$10\%$$

$$f_2 = 5$$

$$\frac{5}{20} = 0.25$$

$$25\%$$

$$f_3 = 5$$

$$\frac{5}{20} = 0.25$$

$$25\%$$

$$f_4 = 4$$

$$\frac{4}{20} = 0.2$$

$$20\%$$

$$f_5 = 3$$

$$\frac{3}{20} = 0.15$$

$$15\%$$

$$f_6 = 1$$

$$\frac{1}{20} = 0.05$$

$$5\%$$

$$\sum f = 20$$

$$\frac{20}{20} = 1$$

$$100\% = 1$$



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Construct the frequency distribution of relative data.

10 9 8 11 7 6 10  
 12 14 20 6 8 6 10  
 8 11 13 7 13 12 15 16  
 17 18 19 20 11 18 19 16

min value of data = 6

max value of data = 20

Range = max - min = 20 - 6 = 14

$k = 1 + 3.33 \log(20)$

$= 5.5 \approx 6.0$

Class width =  $\frac{\text{Max} - \text{min}}{k} = \frac{20 - 6}{6} = \frac{14}{6} = 2.33$   
 $\approx 3 \approx 2$

| Class boundaries | Tally marks | frequency |
|------------------|-------------|-----------|
| 6 less than 8    |             | $f_1 = 5$ |
| 8 less than 10   |             | $f_2 = 4$ |
| 10 less than 12  |             | $f_3 = 5$ |
| 12 less than 14  |             | $f_4 = 4$ |
| 14 less than 16  |             | $f_5 = 2$ |
| 16 less than 18  |             | $f_6 = 4$ |
| 18 less than 20  |             | $f_7 = 4$ |
| 20 less than 22  |             | $f_8 = 2$ |

$\Sigma f = 30$



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Date: / /

| relative freq | % freq   | comulative freq |
|---------------|----------|-----------------|
| $5/20 = 0.16$ | 16%      | 5               |
| $4/20 = 0.13$ | 13.3%    | $5+4=9$         |
| $5/20 = 0.16$ | 16%      | $9+5=14$        |
| $4/20 = 0.13$ | 13.3%    | $14+4=18$       |
| $2/20 = 0.06$ | 6%       | $18+2=20$       |
| $4/20 = 0.13$ | 13.3%    | $20+4=24$ ✓     |
| $4/20 = 0.13$ | 13.3%    | $24+4=28$       |
| $2/20 = 0.06$ | 6%       | $28+2=30$ ✓     |
|               | $= 97.2$ |                 |



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## 2. GRAPHS:

## CHARTS

i. Histogram

i. Bar chart

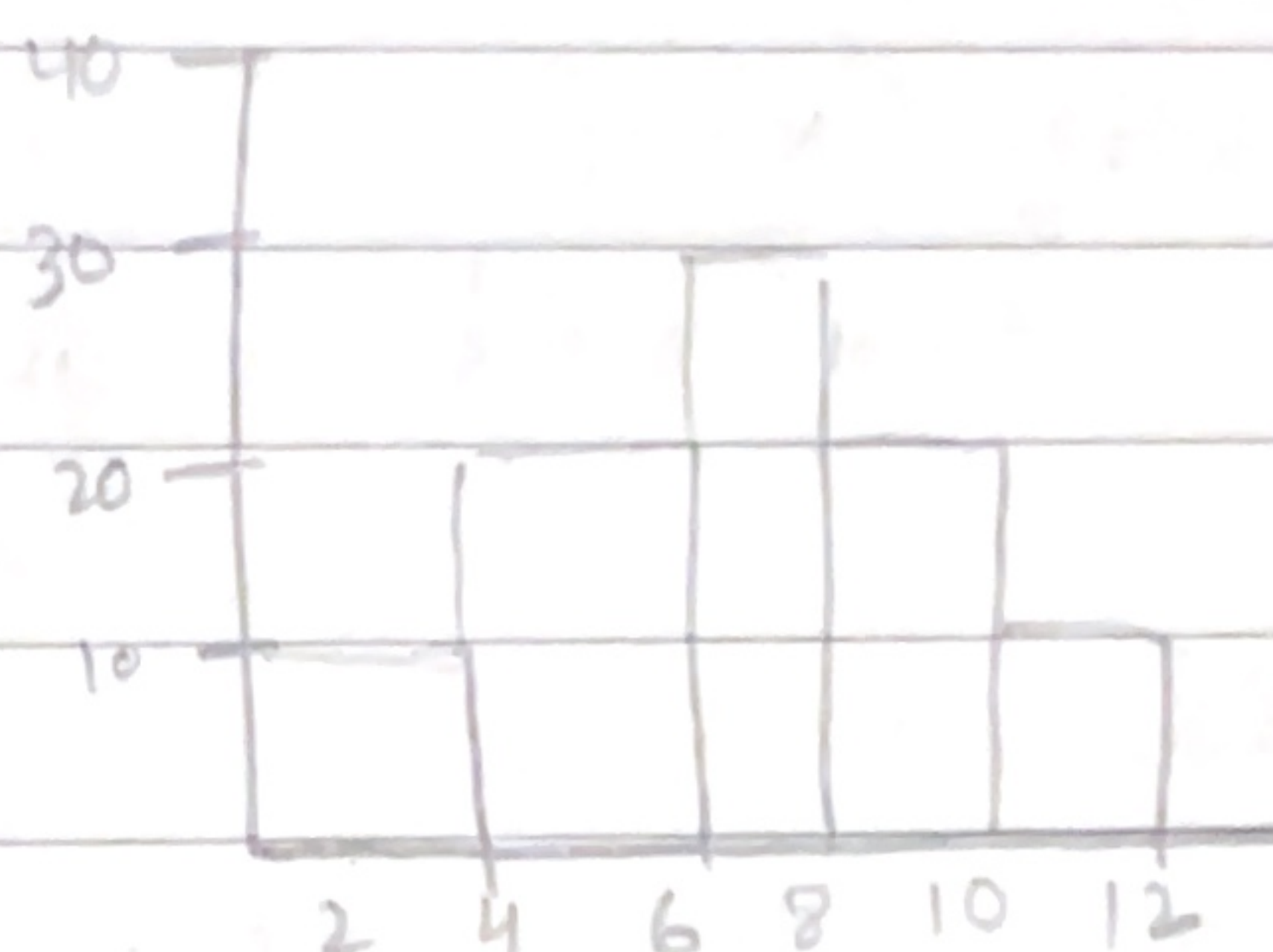
ii. Frequency polygon.

ii. Pie chart

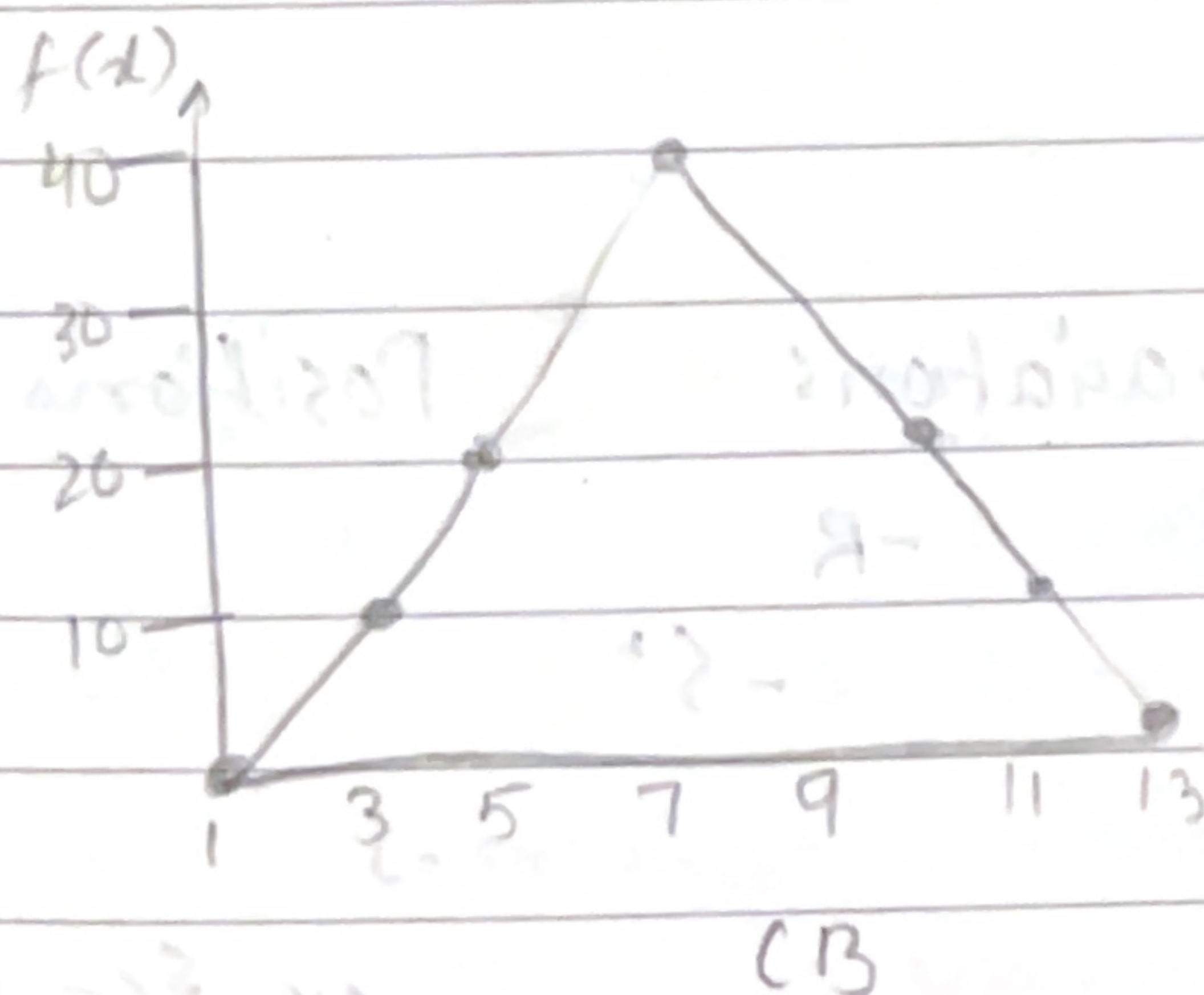
iii. Oblique.

| LB    | Freq | CF  |
|-------|------|-----|
| 2-4   | 10   | 10  |
| 4-6   | 20   | 30  |
| 6-8   | 40   | 70  |
| 8-10  | 20   | 90  |
| 10-12 | 10   | 100 |

$\Sigma f = 100$

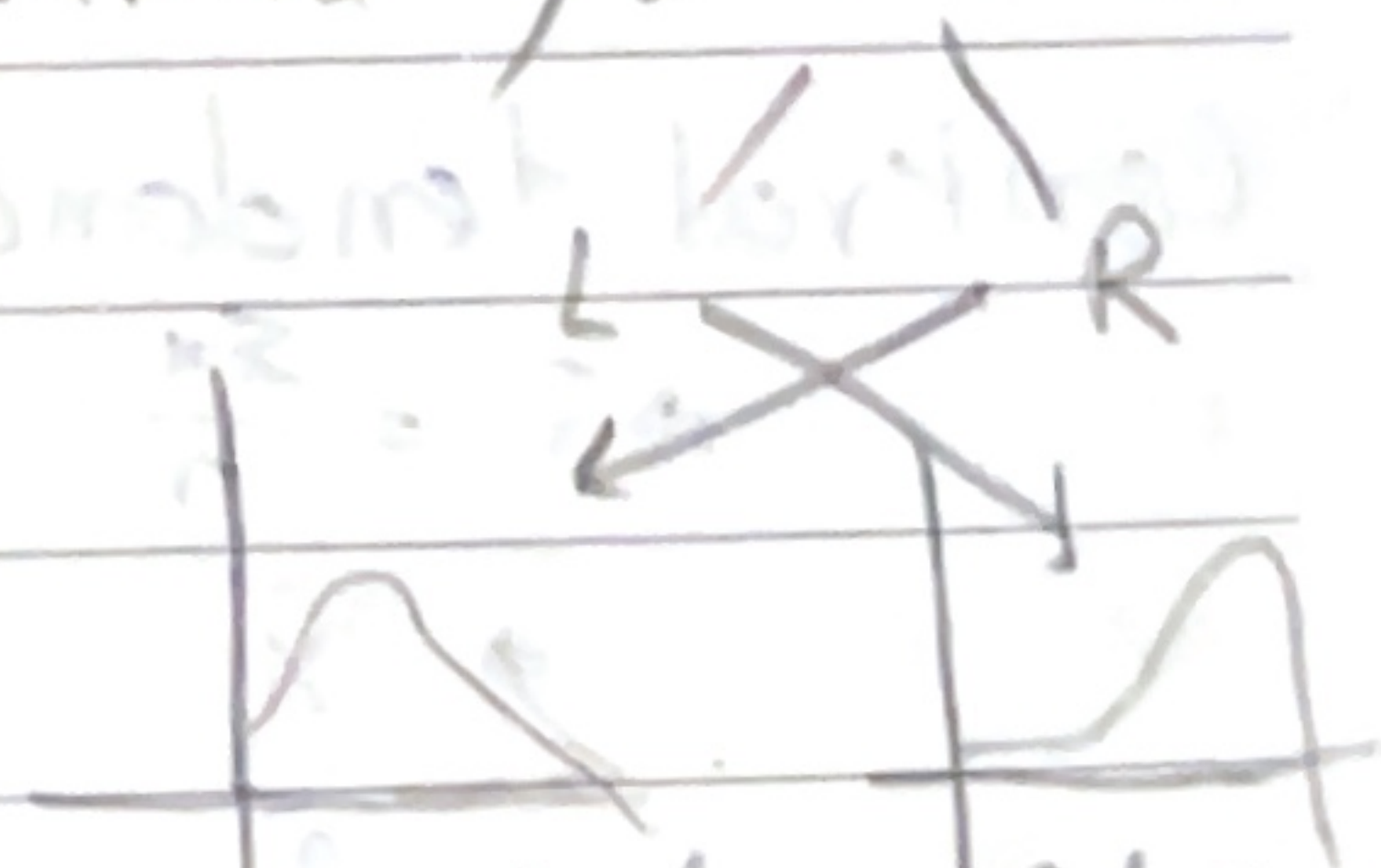


## FREQUENCY POLYGON:



Shape

Normal / skewed



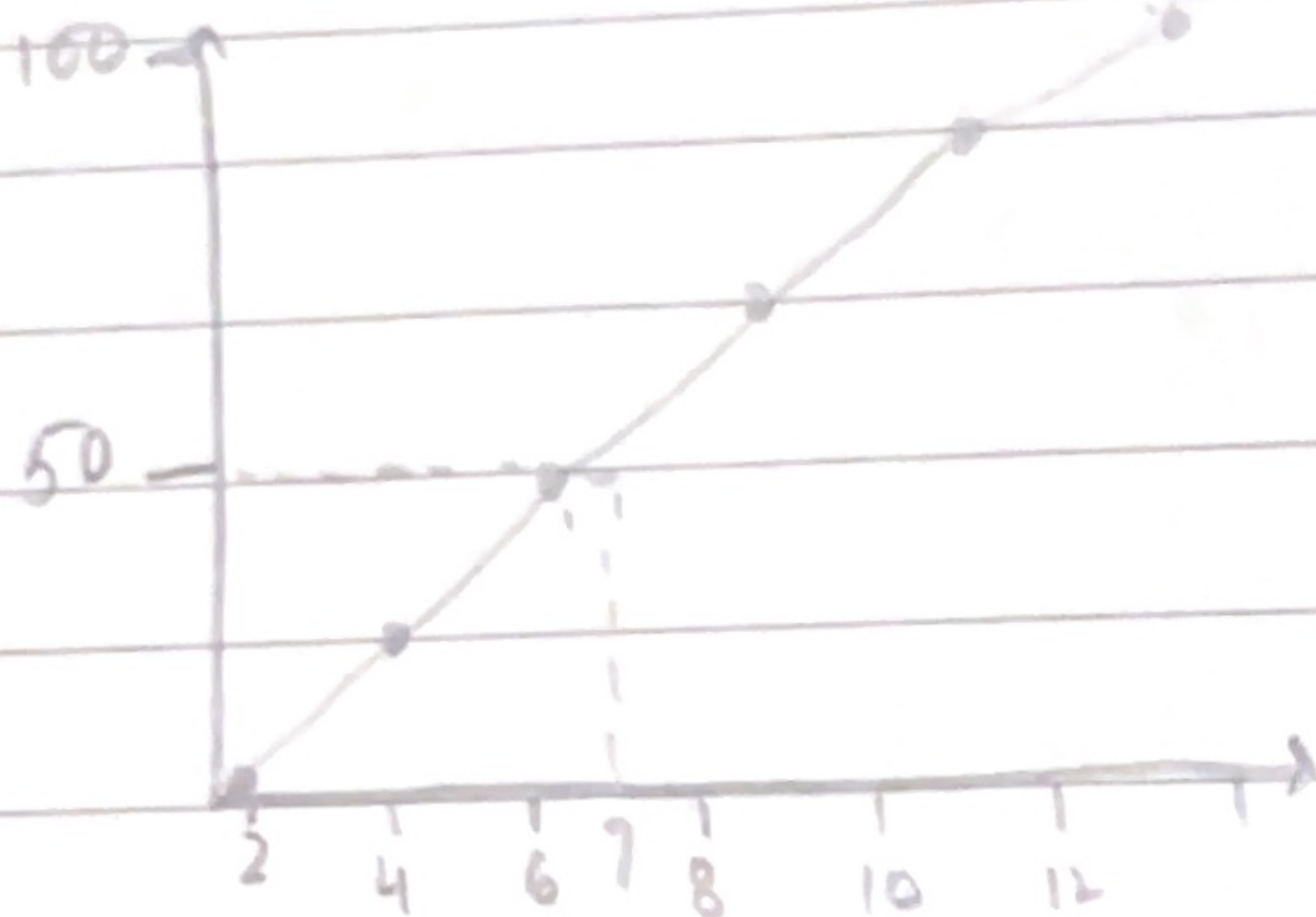
$\Rightarrow$  lack of data = skewed

On Y-axis  $\&$  x-axis gap will be same.



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Ogive:



Midpoint Data

$$\frac{n}{2} = \frac{100}{2} = 50 \text{ midpoint}$$

⇒ Bar chart / Pie chart use for Yes/No data.

Central tendency

Variations

Positions

Mean  $\Rightarrow \bar{x} = \frac{\sum x}{n}$

Range - R

percentiles

Median  $\Rightarrow \tilde{x}$

variance -  $s^2$

deciles

mode  $\Rightarrow \hat{x}$

Standard deviation = S

Quintiles

GM  $\Rightarrow GM$

Co-efficient of variation  $CV = \frac{s}{\bar{x}} \times 100$

HM  $\Rightarrow HM$

" of skewness  $SK = \frac{3(\bar{x} - Med)}{s}$

etc

s



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$n=8$

Sample data  $\Rightarrow$  work done on small data

Population data  $\Rightarrow$  work done on whole bigdata,  
 $n=41$

| Data 1     | Data 2   |
|------------|----------|
| Exercise   | Sleeping |
| time hours | hours    |

|                 |                 |                                                                                                 |
|-----------------|-----------------|-------------------------------------------------------------------------------------------------|
| 2.0             | 6               | Sample mean<br>$\bar{x}_s = \frac{2.0 + 3.0 + \dots + 1.0}{6}$<br>$= 17/6 = 2.83$ hours per day |
| 3.0             | 7               |                                                                                                 |
| 2.0             | 7               |                                                                                                 |
| 4.0             | 6               |                                                                                                 |
| 5.0             | 4               |                                                                                                 |
| 1.0             | 5               |                                                                                                 |
| $\Sigma x = 17$ | $\Sigma y = 35$ | $\bar{x}_s = \frac{6 + 7 + \dots + 5}{6}$                                                       |

Sample  $\Rightarrow n=6$

$= 5.83$  hours.

$\Rightarrow$  MEDIAN (middle most)

• Arrange data in ascending order.

1.0, 2.0, 3.0, 4.0, 5.0

• Pick the middle most value.

$$\frac{2+3}{2} = 2.5$$



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Date: / /

Sleeping hour median.

→ 4 5 6 6 7 7

$$\frac{6+6}{2} = 6 \text{ hours}$$

→ Mode (most repeated value)

$$\hat{x}_c = 2.0 \text{ hours}$$

$$\hat{x}_s = 6, 7 \text{ hours.}$$

→ RANGE (max - min)

$$R_c = 5.0 - 1.0 = 4 \text{ hours}$$

$$R_s = 7 - 4 = 3 \text{ hours.}$$

Sample variance  $\Rightarrow s^2$

$$s^2 = \frac{\sum [x_i - \bar{x}]^2}{n-1}$$

$$s^2 = \frac{(2.0 - 2.83)^2 + (3.0 - 2.83)^2 + \dots + (1.0 - 2.83)^2}{6-1}$$



Date: / /

$$S^2 = \frac{0.6889 + 0.0289 + 0.6889 + 1.3689 + 4.7089 + 3.3489}{5}$$

$$S^2 = \frac{10.8334}{5} = 2.166$$

Taking square root on b.s

$$S = \sqrt{2.166}$$

$$S = 1.46$$

$$CV = \frac{1.46}{2.83} \times 100 = 51.93\%$$

$$SK = \frac{3(2.83 - 2.5)}{1.46} = 0.67$$

(Normal)

It  $\rightarrow SK < +3$  Norm

$SK < -3$  L.S

$SK > +3$  R.S



Date: / /

$$S_s^2 = \frac{(6 - 5.83)^2 + \dots + (5 - 5.83)^2}{6 - 1}$$

$$S_s^2 = \frac{6.081}{5} = 1.216 \text{ hours}^2$$

Taking sq root B.S

$$\sqrt{S_s^2} = \sqrt{1.216}$$

$$S_s = 1.103 \text{ hours}$$

$$CV_s = \frac{1.103}{5.83} \times 100 = 19.09\%$$

$$SK_s = \frac{3(5.83 - 6)}{1.103} = -0.43$$

Normal.  $\therefore -3+3$



Date: / /

3 data

Data 1

Data 2

Data 3

Probability.

Numerical value which describe the possibility of event occurrence.

$$LHS = 30 \quad P(LHS) = \text{Prob. of LHS.}$$

$$P(CS) = 1$$

$$\text{Total} = 50 = \frac{30}{50} = 0.6 \text{ ans } P(MS) = 0.$$

$$0 \leq P(A) \leq 1$$

$$P(\text{Sample Space}) = 1$$

RULES OF PROBABILITY.

1

Simple prob.

Single category

$$P(A) = \frac{\text{NO. of favourable outcomes of an event A.}}{\text{Total no. of outcomes in the sample space.}}$$

Total no. of outcomes in the sample space.



Date: / /

2  
Joint Prob (and)

Two category.

$$P(\text{Suzuki and hybrid}) = \frac{20}{100} = 0.2$$

- Bivariate

|           | Suzuki | Honda | Total |
|-----------|--------|-------|-------|
| Hybrid    | 20     | 40    | 60    |
| nonhybrid | 10     | 30    | 40    |
| Total     | 30     | 70    | 100   |

$$P(\text{Suzuki and nonhybrid}) = \frac{10}{100} = 0.1$$

$$P(\text{Honda and hybrid}) = \frac{40}{100} = 0.4$$

$$P(\text{Honda and nonhybrid}) = \frac{30}{100} = 0.3$$

3) ADDITIVE RULE (OR)  
(A OR B) =

$$P(A) + P(B) - P(A \text{ and } B)$$

P(Honda or Hybrid)



Date: / /

$$= P(H) + P(Hy) - P(H \text{ and } Hy)$$

$$= \frac{40}{100} + \frac{60}{100} - \frac{40}{100}$$

$$= \frac{90}{100} = 0.9$$

$$P(S \text{ or } Hy) = 30/100 + 60/100 - 20/100 = 0.7$$

$$P(S \text{ or non } Hy) = 60/100 + 40/100 - 10/100 = 0.9$$

$$P(H \text{ or } Hy) = 70/100 + 60/100 - 40/100 = 0.9$$

$$P(H \text{ or non } Hy) = 70/100 + 40/100 - 30/100 = 0.8$$

$$P(S \text{ or } H) = 30/100 + 70/100 - 0 = 1$$

→ 1 is the prob of Suzuki or hybrid.

#### 4. CONDITIONAL PROB (given that)

$$P(A \text{ given that } B) =$$

$$P(A/B) = \frac{P(A \text{ and } B)}{P(B)}$$

$$P(S/Hy) = \frac{P(S \text{ and } Hy)}{P(Hy)} \quad \because 0.3 \text{ is probability of Suzuki hybrid.}$$

$$P(S/Hy) = \frac{20/100}{60/100} = \frac{1}{3} = 0.3$$



Date: / /

$$P(S/NHy) = \frac{P(S \text{ and } NHy)}{P(NHy)}$$

$$= \frac{0.1}{0.4} = 0.25$$

$$P(H/Hy) = \frac{P(H \text{ and } Hy)}{P(Hy)}$$

$$= \frac{0.4}{0.6} = 0.66$$

$$P(H/NHy) = \frac{P(H \text{ and } NHy)}{P(NHy)}$$

$$= \frac{0.3}{0.4} = 0.75$$

50 Complement-

$$\overset{\text{hybrid}}{P(\text{NOT interested})} = 1 - \overset{\text{non hybrid}}{P(\text{interested})}$$

$$P(RHS) = 1 - P(LHS)$$

$$= 1 - 0.6$$

$$= 0.4 \text{ Ans}$$



Date: / /

# POISSON PROBABILITY DISTRIBUTION

1. Time per hour, per month etc
2. Space, in parking, per page, etc
3. Volume, tank, bottle, etc.

mean  
(avg)  
 $= \lambda$

$x$  = No. of call recieved per hour.

$$P(x=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Example:

$x$  = no. of car washed at a store per day

$\mu = 5.5$  per day

If we select a day at random, find the prob that no. of cars washed would be.



Date: / /

- a)  $P(\text{less than } 2) = P(0) + P(1)$   
 b)  $P(\text{at most } 3) = P(0) + P(1) + P(2) + P(3)$   
 c)  $P(\text{at least } 1) = P(1) + P(2) + \dots$   
 d)  $P(\text{more than } 4) = P(5) + P(6) + \dots$   
 e)  $P(\text{exactly } 5) = P(5)$   
 f)  $P(\text{from } 2 \text{ to } 5) = P(2) + \dots + P(5)$

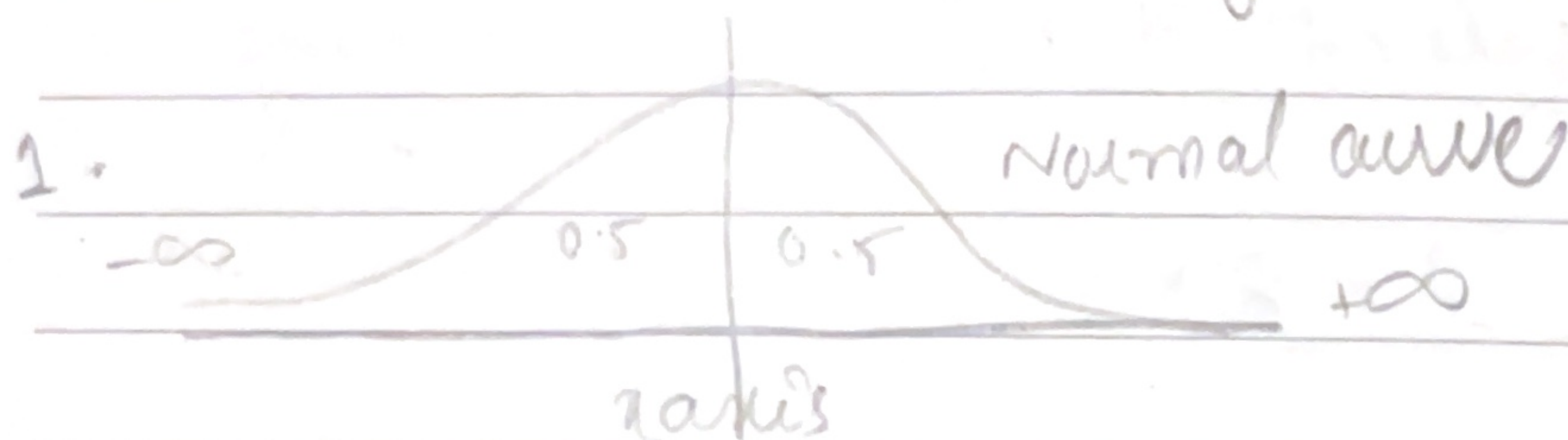
| $x$ | $\lambda = 5.5$ | a) $P(\text{less than } 2)$   |
|-----|-----------------|-------------------------------|
| 0   | .0041           | $0.0041 + 0.0025$             |
| 1   | .0225           | $= 0.0266$                    |
| 2   | .0618           | b) $0.0041 + 0.0225 + 0.0618$ |
| 3   | .1133           | $+ 0.1133 = 0.2017$           |
| 4   | .1558           | c) $0.9958$                   |
| 5   | .1714           | d) $0.4924$                   |
| 6   | .1571           | e) $0.1714$                   |
| 7   | .1234           | f) $0.5023$                   |
| 8   | .0849           |                               |
| 9   | .0519           |                               |
| 10  | .0285           |                               |
| 11  | .0143           |                               |
| 12  | .0065           |                               |
| 13  | .0028           |                               |
| 14  | .0011           |                               |
| 15  | .0004           |                               |
| 16  | .0001           |                               |
| 17  | .0000           |                               |



Date: / /

# NORMAL PROB. DISTRIBUTION.

$X =$  Continuous. kg, height,



2. mean = median = mode

3. Area under the normal curve = 1

4. Z table.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < +\infty$$

$$\text{Area} = \int_{-\infty}^{+\infty} f(x) dx = 1$$

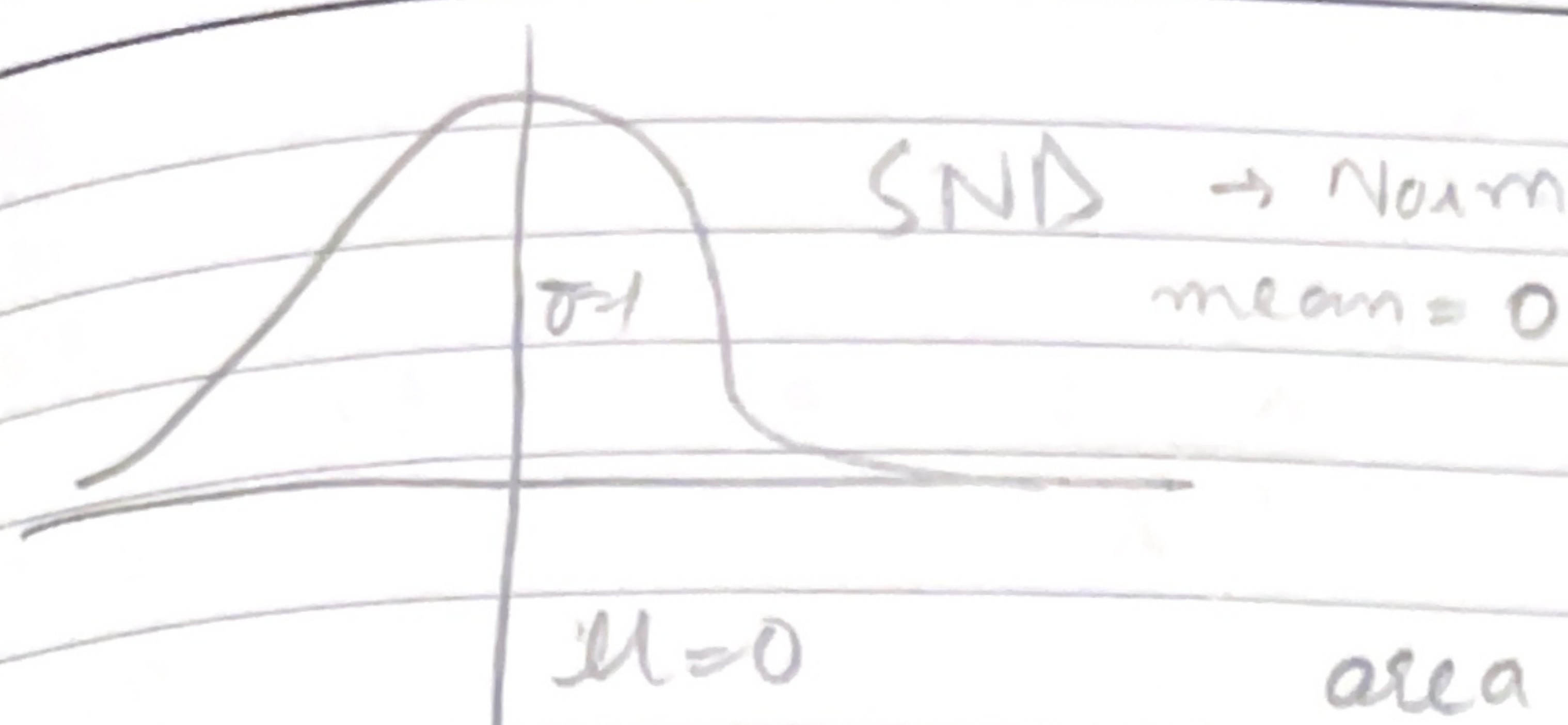
NPD  $\xrightarrow{\text{convert}}$  Standard Normal dist.

$$Z = \frac{x - \mu}{\sigma}$$

Z table



Date: / /



area 1 - area 2  
remaining area  
value will be.

$$\begin{aligned} P(-2.4 < Z < 1.46) &= ? \\ &= P(1.46) - P(-2.40) \\ &= 0.9279 - 0.0082 \\ &= 0.9179 \end{aligned}$$

It is symmetric to mean so  
if mean is zero all three will  
be zero.

Example:

$X$  = weight of parcel (in kg)

$$\mu = 3.5 \quad \sigma = 1.2$$

If we select a parcel randomly,  
find the prob. that weight of  
parcel would be:

- $P(\text{less than } 2\text{kg}) =$
- $P(\text{more than } 5\text{kg}) =$
- $P(\text{b/w } 2 \text{ to } 4\text{kg}) =$

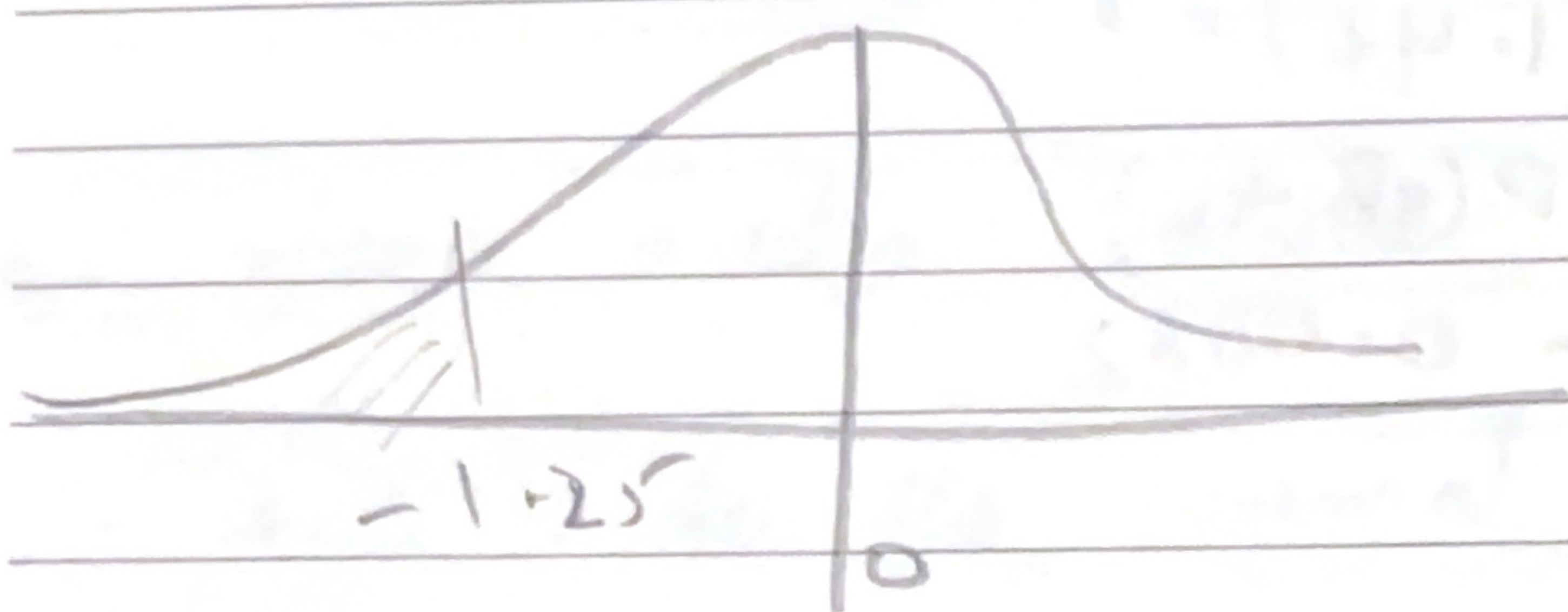


Date: / /

a)  $P(X < 2) =$

$$Z = \frac{x - \mu}{\sigma} = \frac{2 - 3.5}{1.2} = -1.25$$

$$P(Z < -1.25) = 0.1056$$



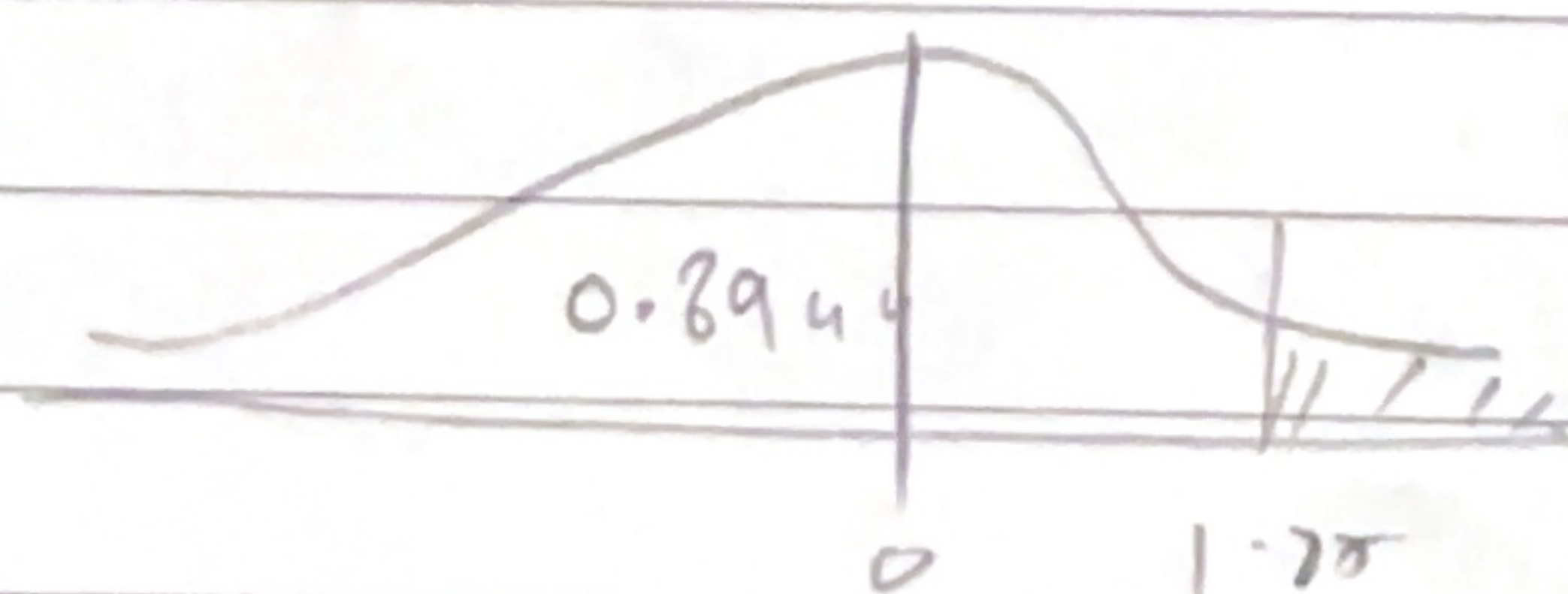
$$P(\text{less than } 2 \text{ kg}) = 0.1056$$

Ans

b)  $P(\text{more than } 5 \text{ kg})$

$$Z = \frac{x - \mu}{\sigma} = \frac{5 - 3.5}{1.2} = +1.25$$

$$P(Z > 1.25) = 0.8944 - 1 = 0.1056$$





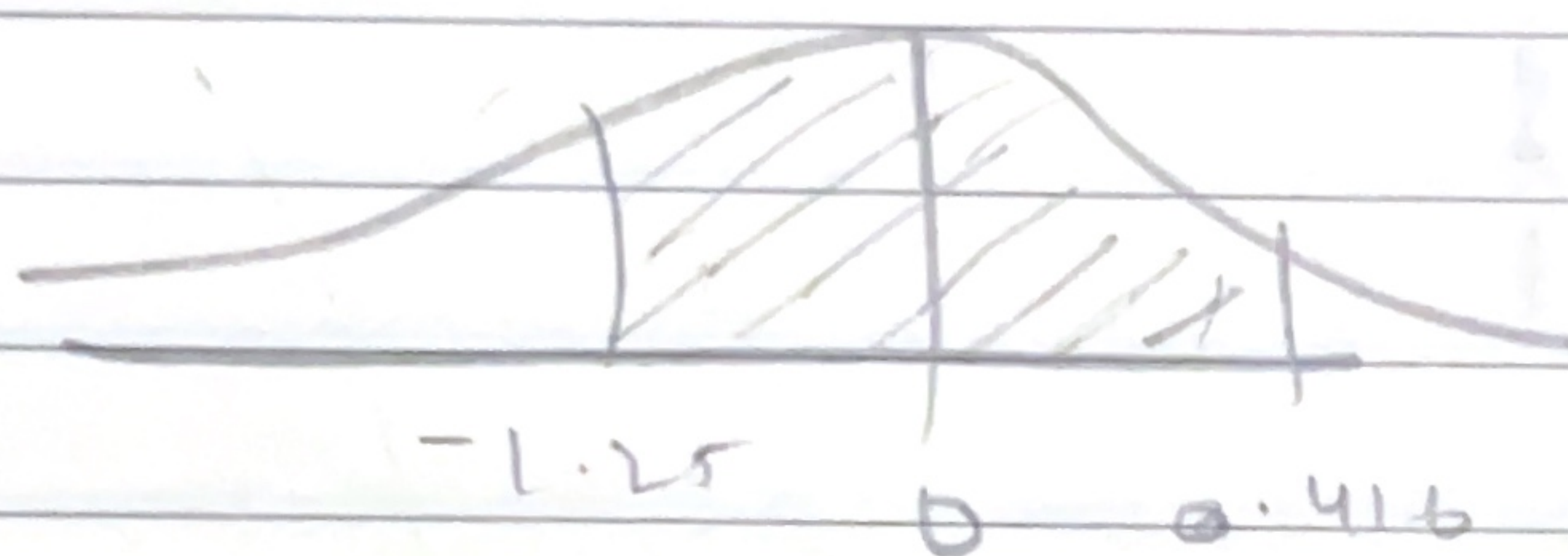
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$$P(X > 5) = 0.1056$$

c)  $P(\text{between } 2 \text{ to } 4 \text{ kg})$

$$Z_1 = \frac{(2 - 3.5)}{1.2} = -1.25$$

$$Z_2 = \frac{(4 - 3.5)}{1.2} = 0.416$$



$$P(Z < -1.25) = 0.1056$$

$$P(Z < 0.41) = 0.6591$$

$$= 0.6591 - 0.1056$$

$$= 0.5535$$

Ans,



Date: / /

Draw normal distribution curve & shade that-derived area that represents the probability. Since the test score are normally distributed, the test value  $x$

334 • 332 • 330 (imp)

• APPROXIMATION:

$$Np \geq 5$$

(353)

$$Np > 5$$

more than  $\Rightarrow +1$

at least  $= -1$